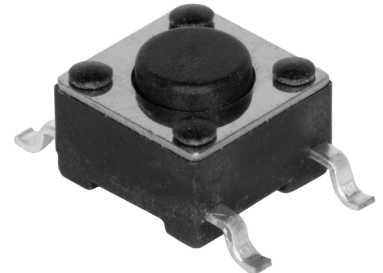


**SERIES:** TS04 | **DESCRIPTION:** TACTILE SWITCH**FEATURES**

- surface mount
- black actuator

**SPECIFICATIONS**

| parameter             | conditions/description                                                                                    | min  | typ                         | max  | units                      |
|-----------------------|-----------------------------------------------------------------------------------------------------------|------|-----------------------------|------|----------------------------|
| rated voltage         |                                                                                                           | 1    |                             | 12   | Vdc                        |
| rated current         |                                                                                                           | 0.01 |                             | 50   | mA                         |
| withstanding voltage  | for 1 minute                                                                                              |      | 250                         |      | Vac                        |
| contact resistance    | applying load of 2 times operating force at 5 Vdc, 10 mA                                                  |      |                             | 100  | mΩ                         |
| insulation resistance | at 100 Vdc for 1 minute                                                                                   | 100  |                             |      | MΩ                         |
| operating force       | see force table                                                                                           |      |                             |      |                            |
| actuator travel       |                                                                                                           | 0.15 | 0.25                        | 0.35 | mm                         |
| operating temperature |                                                                                                           | -30  |                             | 80   | °C                         |
| storage temperature   |                                                                                                           | -40  |                             | 90   | °C                         |
| life                  | rated load, 2 cycles/second, 1.5 times operating force<br>100 gf models<br>160 gf models<br>260 gf models |      | 100,000<br>80,000<br>60,000 |      | cycles<br>cycles<br>cycles |
| vibration             | 10~55~10 Hz, 1.5 mm amplitude, 2 hours on each XYZ                                                        |      |                             |      |                            |
| flammability rating   | see material table                                                                                        |      |                             |      |                            |
| RoHS                  | yes                                                                                                       |      |                             |      |                            |

**PART NUMBER KEY****TS04 - 66 - XX - BK - XXX - SMT - XX**

Base Number

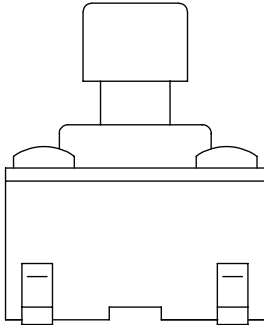
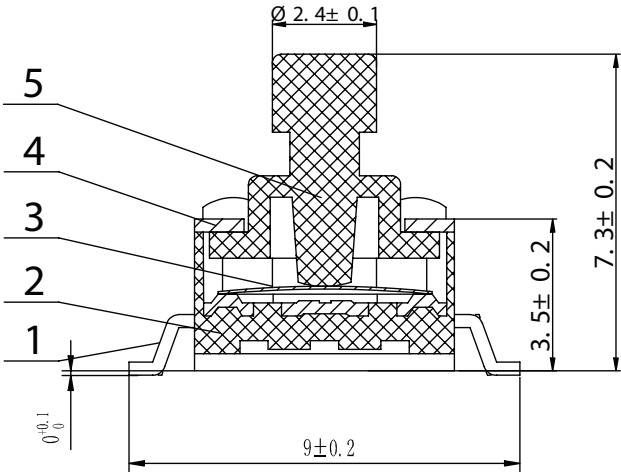
Actuator Height above PCB:  
43 = 4.3 mm    \*73 = 7.3 mm  
50 = 5.0 mm    75 = 7.5 mm  
55 = 5.5 mm    85 = 8.5 mm  
65 = 6.5 mm    95 = 9.5 mm  
70 = 7.0 mmPush Force:  
100 = 100 gf  
160 = 160 gf  
260 = 260 gfPackaging:  
"blank" = bag  
TR = tape and reel

Note: \*7.3 mm height versions have square actuator top. All other models have round actuator top.

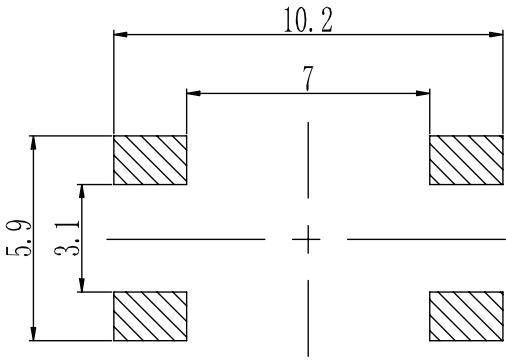
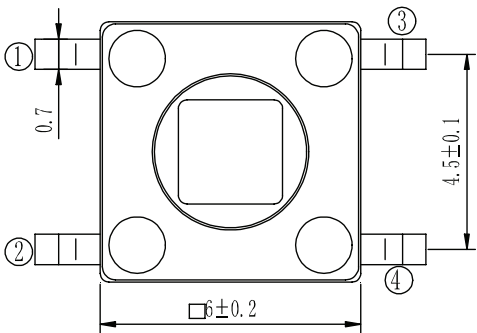
MECHANICAL DRAWING (7.3 MM HEIGHT MODELS)

units: mm  
tolerance: ±0.2 mm  
unless otherwise noted

| ITEM | DESCRIPTION | MATERIAL        | PLATING/COLOR |
|------|-------------|-----------------|---------------|
| 1    | terminals   | brass           | silver        |
| 2    | case        | PA66 (UL94HB)   | black         |
| 3    | contacts    | stainless steel | silver        |
| 4    | cover       | stainless steel |               |
| 5    | stem        | PA66 (UL94HB)   | black         |



| ACTUATOR POSITION<br>[ ]: MOMENTARY |      |
|-------------------------------------|------|
| NORMAL                              | DOWN |
| OFF                                 | [ON] |
| SCHEMATIC                           |      |
| ①                                   | ③    |
| ②                                   | ④    |



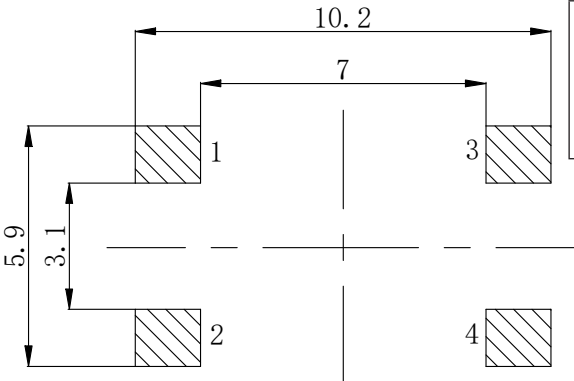
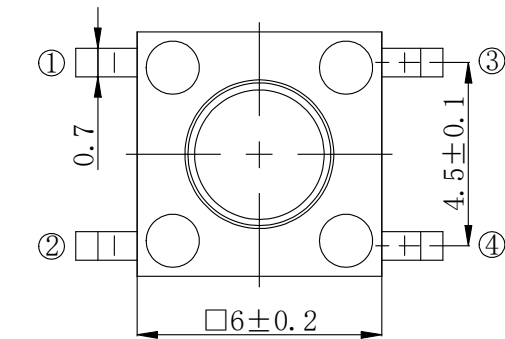
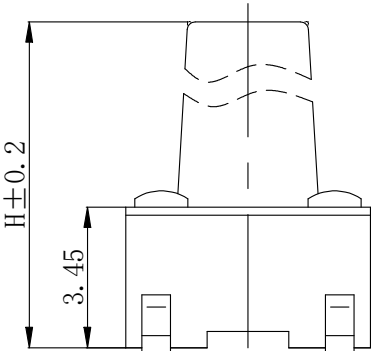
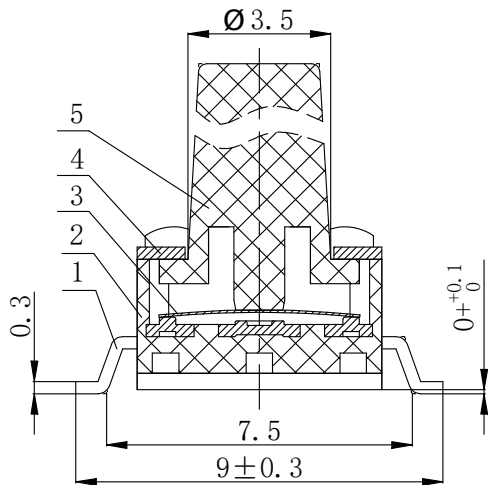
Recommended PCB Layout  
Top View

| Operting Force Table |                 |                  |
|----------------------|-----------------|------------------|
| Model                | Press Force [N] | Return Force [N] |
| 100                  | $1.0 \pm 0.3$   | $> 0.2$          |
| 160                  | $1.6 \pm 0.5$   | $> 0.2$          |
| 260                  | $2.6 \pm 0.5$   | $> 0.2$          |

MECHANICAL DRAWING (ALL OTHER HEIGHT MODELS)

units: mm  
tolerance: ±0.2 mm  
unless otherwise noted

| ITEM | DESCRIPTION | MATERIAL        | PLATING/COLOR |
|------|-------------|-----------------|---------------|
| 1    | terminals   | brass           | silver        |
| 2    | case        | PA66 (UL94HB)   | black         |
| 3    | contacts    | stainless steel | silver        |
| 4    | cover       | stainless steel |               |
| 5    | stem        | PA66 (UL94HB)   | black         |



| ACTUATOR POSITION<br>[ ]: MOMENTARY |      |
|-------------------------------------|------|
| NORMAL                              | DOWN |
| OFF                                 | [ON] |
| SCHEMATIC                           |      |
| ①                                   | ③    |
| ②                                   | ④    |

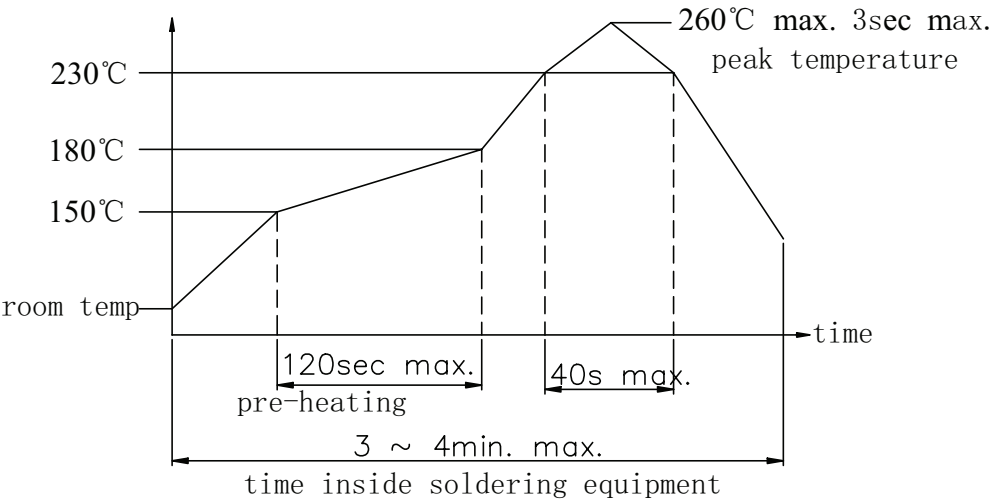
Recommended PCB Layout  
Top View

| Operting Force Table |                 |                  |
|----------------------|-----------------|------------------|
| Model                | Press Force [N] | Return Force [N] |
| 100                  | 1.0±0.3         | >0.2             |
| 160                  | 1.6±0.5         | >0.2             |
| 260                  | 2.6±0.5         | >0.2             |

SOLDERABILITY

| parameter                     | conditions/description | min | typ | max | units |
|-------------------------------|------------------------|-----|-----|-----|-------|
| hand soldering                | for maximum 3 seconds  |     |     | 350 | °C    |
| reflow soldering <sup>1</sup> | see reflow profile     |     |     | 260 | °C    |

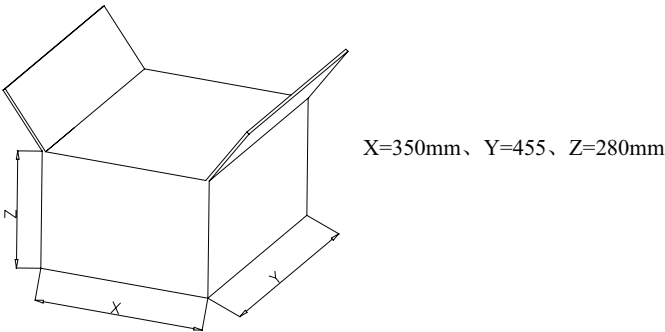
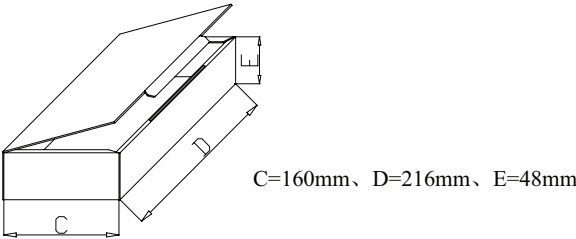
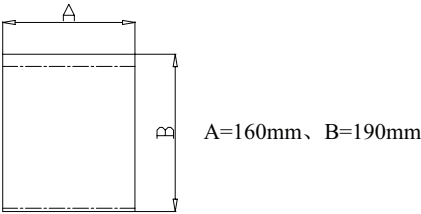
Notes: 1. Do not exceed 2 reflow cycles.



PACKAGING (BAG)

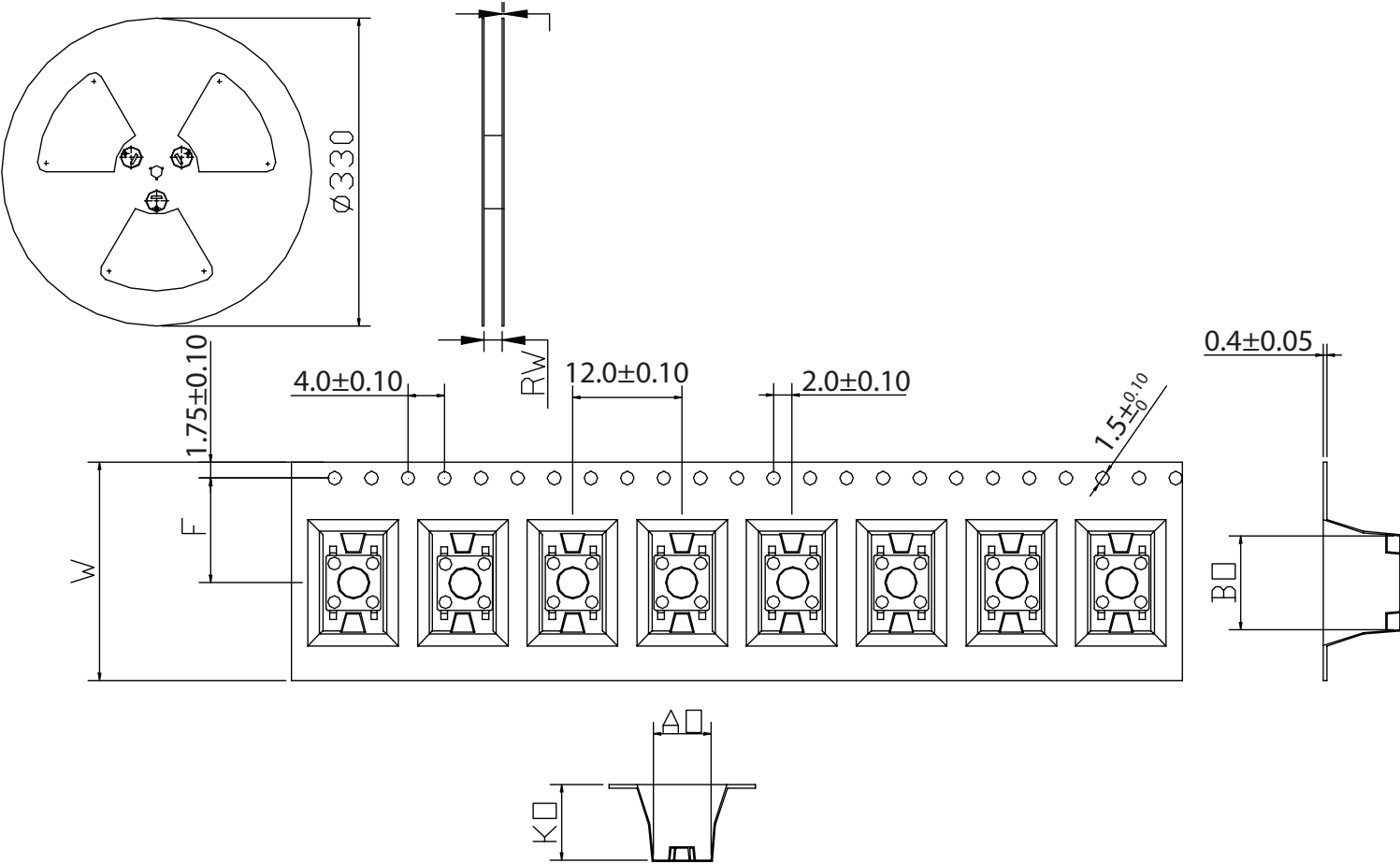
units: mm

Bag Size: 190 x 160 mm  
Carton Box Size: 445 x 350 x 280 mm  
Bag QTY: 1,000 pcs per bag  
Carton Box QTY: 60,000 pcs per carton



PACKAGING (TAPE & REEL)

| Packaging Table             |          |                          |            |         |        |         |          |                                   |          |
|-----------------------------|----------|--------------------------|------------|---------|--------|---------|----------|-----------------------------------|----------|
| Actuator Height Models (mm) | Reel QTY | Carton Size [XxYxZ] (mm) | Carton QTY | RW (mm) | W (mm) | AO (mm) | BO (mm)  | KO (mm)                           | F (mm)   |
| 4.3                         | 1,000    | 340 x 430 x 340          | 20,000     | 16.5    | 16±0.3 | 6.3±0.1 | 10.4±0.1 | 4.5±0.1                           | 7.5±0.1  |
| 5.0                         | 1,000    | 340 x 430 x 340          | 20,000     | 16.5    | 16±0.3 | 6.3±0.1 | 10.4±0.1 | 5.2±0.1                           | 7.5±0.1  |
| 5.5                         | 1,000    | 340 x 430 x 340          | 20,000     | 16.5    | 16±0.3 | 6.3±0.1 | 10.4±0.1 | 5.7±0.1                           | 7.5±0.1  |
| 6.5                         | 800      | 340 x 430 x 340          | 16,000     | 16.5    | 16±0.3 | 6.3±0.1 | 10.4±0.1 | 6.7±0.1                           | 7.5±0.1  |
| 7.0                         | 750      | 340 x 430 x 340          | 10,500     | 24.5    | 24±0.3 | 6.3±0.1 | 10.3±0.1 | 7.3±0.1                           | 11.5±0.1 |
| 7.3                         | 750      | 340 x 430 x 340          | 10,500     | 24.5    | 24±0.3 | 6.4±0.1 | 10.5±0.1 | 7.8±0.1                           | 11.5±0.1 |
| 7.5                         | 750      | 340 x 430 x 340          | 10,500     | 24.5    | 24±0.3 | 6.4±0.1 | 10.5±0.1 | 7.8±0.1                           | 11.5±0.1 |
| 8.5                         | 500      | 340 x 430 x 340          | 7,000      | 24.5    | 24±0.3 | 6.3±0.1 | 10.4±0.1 | 8.6 <sup>+0.1</sup> <sub>-0</sub> | 11.5±0.1 |
| 9.5                         | 500      | 340 x 430 x 340          | 7,000      | 24.5    | 24±0.3 | 6.3±0.1 | 10.4±0.1 | 9.6 <sup>+0.1</sup> <sub>-0</sub> | 11.5±0.1 |



Note: 1. Actuator height 4.3 & 5.0 models have about 15~25 open positions reserved for the front and rear positions of the reel.  
2. Actuator height 5.5 & 6.5 models have about 15~30 open positions reserved for the front and rear positions of the reel.  
3. Actuator height 7.0, 7.3, & 7.5 models have about 20~35 open positions reserved for the front and rear positions of the reel.  
4. Actuator height 8.5 & 9.5 models have about 30~50 open positions reserved for the front and rear positions of the reel.

REVISION HISTORY

| rev. | description                                          | date       |
|------|------------------------------------------------------|------------|
| 1.0  | initial release                                      | 09/20/2021 |
| 1.01 | added 7.3 mm height model drawing                    | 11/03/2021 |
| 1.02 | logo, datasheet style update                         | 08/05/2022 |
| 1.03 | added tape and reel packaging option                 | 05/12/2023 |
| 1.04 | CUI Devices rebranded to Same Sky                    | 09/12/2024 |
| 1.05 | modified terminal shape and changed terminal plating | 11/03/2025 |

The revision history provided is for informational purposes only and is believed to be accurate.



Same Sky offers a one (1) year limited warranty. Complete warranty information is listed on our website.

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